Workout 1A Homework 1 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:09

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

A set of 27 numbers which are spaced evenly apart have a sum of 108. What is the value of the number that is in the middle?

2

Review Later

The sum of the first 2 odd numbers is equal to $1 + 3 = 4$.

The sum of the first 3 odd numbers is equal to $1 + 3 + 5 = 9$.

The sum of the first 4 odd numbers is equal to $1 + 3 + 5 + 7 = 16$.

Following this pattern, the sum of the first 10 odd numbers is equal to which of the following?

- 9^2
- 10^2
- 12^2
- 8^2
- 11^2
- 7^2

3

Review Later

In shoe sizing, a given number shoe size for a women's shoe is actually a different size from the same number size for a men's shoe. The chart below shows a size conversion between women's and men's shoes. Let w equal a women's shoe size, and m the equivalent size in men's shoes. What is the relationship between w and m?

Women's Sizing	7.5	8	8.5	9	9.5
Men's Sizing	6	6.5	7	7.5	8

- $m = 1.5 + w$
- $w = 1.5 + m$
- $0.5w = 1.5m$
- $w = 1.5m$
- $m = 1.5w$
- $w = 0.5 + m$

4

Review Later

Back in the time of the Romans, shoes were originally measured in terms of a unit called barley corns. Instead of barley corns, we now define a unit of shoe size to be 1 centimeter in every dimension. What percent is the volume of a size 8.5 foot 24.5 cm long compared with a size 9 foot measuring 25 cm long? Choose the closest answer.

- 94.4%
- 86.8%
- 82.4%
- 99.7%
- 91.5%

5

Review Later

For Halloween, a pastry chef bakes pumpkin cookies and skeleton cookies. The pumpkin cookies are 3 inches long and 3.5 inches wide. The skeleton cookies are 4 inches long and 1.5 inches wide. If the pastry chef was able to fit 96 pumpkin cookies on his cookie sheets, how

many skeleton cookies can he fit on all of his cookie sheets? Your answer should be the number only, without units.

6

Review Later

Suppose you started with a single-spaced 6-page report in single-spaced 8-point font and switched to 12-point double-spaced font. How many pages will your report end up being?

- 12 pages
- 9 pages
- 27 pages
- 18 pages
- 16 pages
- 24 pages
- 30 pages

7

Review Later

Evan averages 88 points on his five tests. The average of his two top scores is 92, and the average of his two low scores is 85. What was his middle score?

8

Review Later

On a distant planet, two-thirds of the aliens have six legs and the rest have four legs. If there are 2560 legs on the planet, then how many aliens are on the planet?

9

Review Later

Sam eats twice as many gummy bears as Vishanth, and one-third as many gummy bears as Pei. If Pei eats 90 more gummy bears than Vishanth, then how many gummy bears were eaten altogether?

10

Review Later

What is $\frac{17+19+21+23+25+\ldots+151+153}{15}$?

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

The sum of 4 consecutive odd integers is 56. What is the smallest number?

12

Review Later

Compare $\frac{\sqrt{24}}{\sqrt{5}}$ and $\frac{\sqrt{75}}{\sqrt{18}}$.
Which one is bigger?

- $\frac{\sqrt{75}}{\sqrt{18}}$
- $\frac{\sqrt{24}}{\sqrt{5}}$
- They are the same

13

Review Later

Approximately how many digits are there in the number $\frac{7^{77}}{5^{37}}$?

digits

14

Review Later

Suppose that in 4 minutes, 6 experienced kids can capture 3 flags and 13 inexperienced kids can capture 4 flags. How many minutes will it take 2 inexperienced and 3 experienced kids to capture 55 flags?

15

Review Later

How many different solutions (m, n) are there that satisfy $2^m = 9^n$ for some integers m and n?

16

Review Later

What is the value of $7(2^0) + 6(2^1) + 5(2^2) + 4(2^3) + 3(2^4) + 2(2^5) + 2^6$?

17

Review Later

If $\dfrac{2020^2 - a^2}{2017^2 - b^2} = 7$ and a is an integer, what is the value of a?

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

Find the sum of all three-digit integers such that 13 times the sum of their digits equals the number itself.

19

Review Later

What is the value of
$2^0 + 2(2^1) + 3(2^2) + 4(2^3) + 5(2^4) + 6(2^5) + 5(2^6) + 4(2^7) + 3(2^8) + 2(2^9) + 2^{10}$?

20

Review Later

What is the value of $2^0 + 3(2^1) + 6(2^2) + 7(2^3) + 6(2^4) + 3(2^5) + 2^6$?

View Marked Questions

Finished

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